Iteration 5

CIS 320-01

Team: Louisville Codes

Spencer Taylor, Gerrit Kloek, Logan Riedley, Hunter Watring

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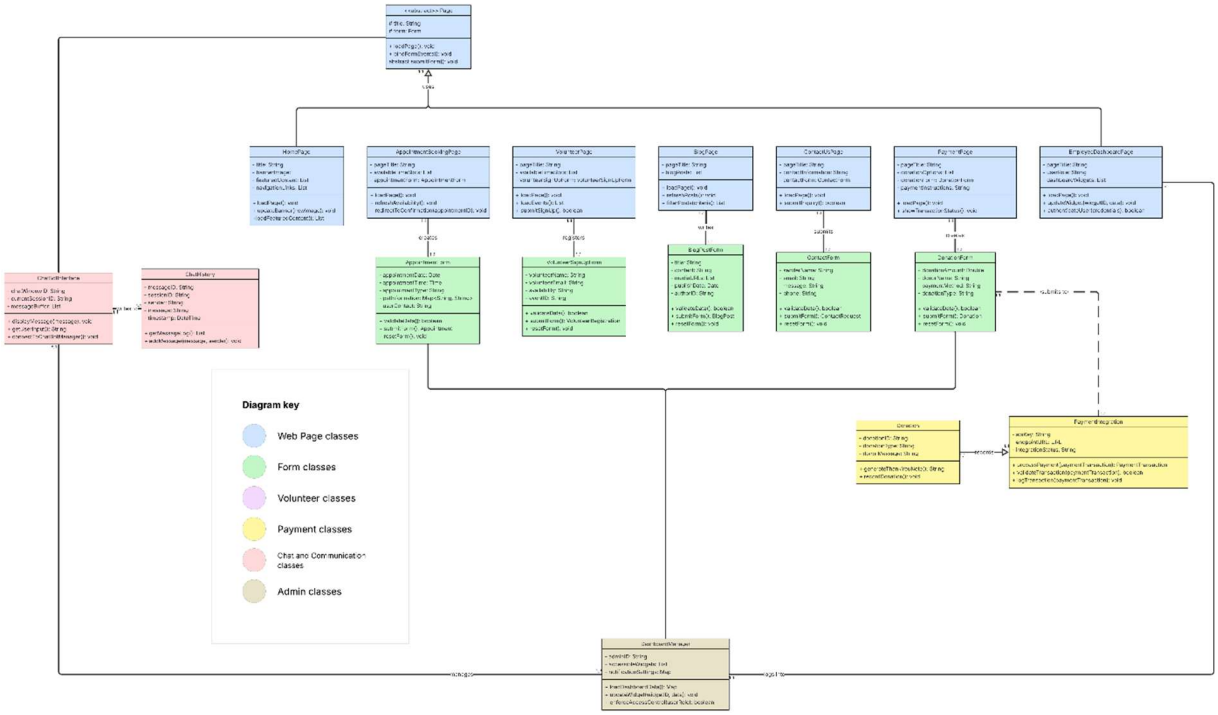
1. Class Diagram

1.1 Class Diagram Narrative

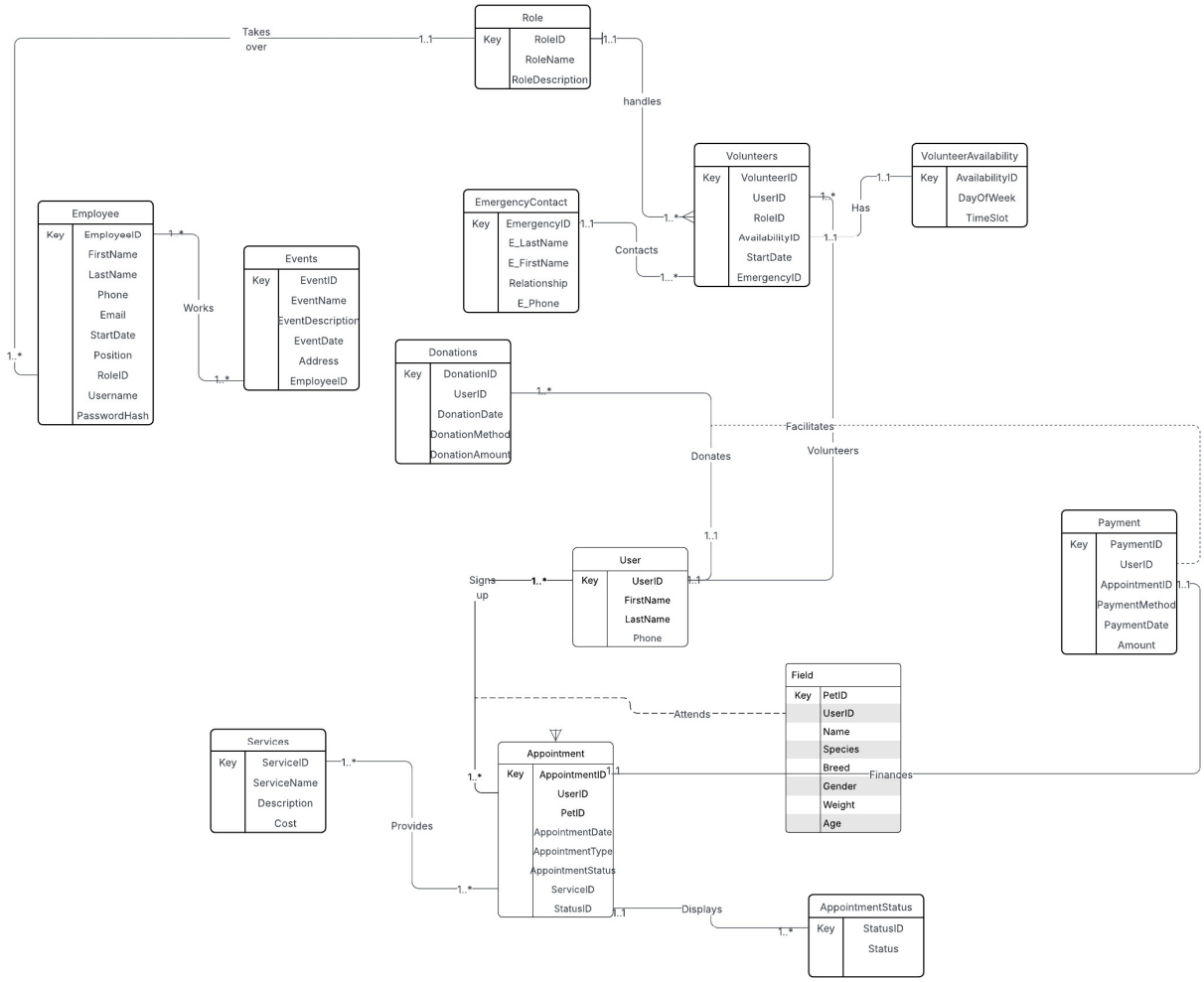
Our class diagram outlines all the major structural components needed to support the Pets4Life Louisville website. The system is broken down into several categories, each representing a specific part, with color-coded classes. The blue classes represent the pages a user sees, they all inherit from the abstract page class, which contains shared attributes like the page title and the form being shown. Concrete page types such as AppointmentBookingPage, VolunteerPage, and PaymentPage each include page-specific attributes and operations (like refreshAvailability() for appointments or authenticateUser() for employee login). The green classes are the forms associated with each use case, such as AppointmentForm, VolunteerSignUpForm, and DonationForm. These forms deal with user inputs, validate them, and submit it to generate a new object (such as a Donation or Appointment). The yellow section represents the core elements related to online payments. The DonationForm class submits to the PaymentIntegration class, which handles the payment logic and ultimately creates a Donation object. The donation and payment handling flow is especially as it ensures both secure and organized processing of payments. The pink classes on the left are part of the chatbot system. ChatBotInterface manages the user interaction, and ChatHistory logs everything through. These pieces work together to provide support or information to users interacting with the website. Finally, the beige DashboardManager sits at the bottom center as a coordinating class for all administrative functions. It allows authorized users to view manage information on pages (such as creating posts or new events) and enforce access control which restricts the page to only authorized users.

1.2 Class Diagram



2. Database Diagram

2.1 Database Diagram



2.2 Database Diagram Narrative

The database was designed to support the organization by efficiently managing user data. The user table stores data regarding each user that donates or makes an appointment within the website. The appointment table stores necessary information from other tables such as the pet and services table that would be relevant to the appointment itself. The payment table was made specifically for appointment payments while the donations table has its own separate table. Employee and volunteer information is stored within the employee and volunteer tables respectively, and each are tied to the roles table which will specify the position/role of that employee/volunteer. The volunteer availability table and the emergency contact table are both exclusively tied to the volunteer table. The events table houses relevant information regarding scheduled events and is tied to the EmployeeID of that who scheduled the event.

2.3 Database Definitions

Client

Attribute	Definition	Type	Size	Key
ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Primary
FirstName	User's first name	String	255 bytes(Max)	No
LastName	User's last name	String	255 bytes(Max)	No
Phone	User's phone number	String	30 bytes	No
Email	User's email address	String	255 bytes(max)	No
PetID	Unique identifier for each pet	Unsigned Long	8 bytes	Foreign

Appointment

Attribute	Definition	Type	Size	Key
AppointmentID	Unique identifier for each appointment	Unsigned Long	8 bytes	Primary
ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Foreign
PetID	Unique identifier for each pet	Unsigned Long	8 bytes	Foreign
AppointmentDate	Date of the scheduled appointment	Date	3 bytes	No
ServiceTypeID	Unique identifier for each service type	Int	4 bytes	Foreign
AppointmentStatus	Status of appointment	String	255 bytes(Max)	No

Pet

Attribute	Definition	Type	Size	Key
PetID	Unique identifier for each pet	Unsigned Long	8 bytes	Primary
ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Foreign
Name	Pet's name	String	255 bytes(Max)	No
Species	Species of animal	String	255 bytes(Max)	No
Breed	Pet breed	String	255 bytes(Max)	No
Gender	Gender of the pet	String	10 bytes	No

Weight	Weight of pet in pounds	Float	4 bytes	No
Age	Age of pet in years	Int	4 bytes	No

Payment

Attribute	Definition	Type	Size	Key
PaymentID	Unique identifier for each payment	Unsigned Long	8 bytes	Primary
ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Foreign
AppointmentID	Unique identifier for each Appointment	Unsigned Long	8 bytes	Foreign
PaymentMethod	Method of payment	String	255 bytes(Max)	No
PaymentDate	Date of payment	Date	3 bytes	No
PaymentAmount	Total amount paid in USD	Double	8 bytes	No

Services

Attribute	Definition	Type	Size	Key
ServiceID	Unique identifier for each service	Int	4 bytes	Primary
ServiceName	Name of service	String	255 bytes(Max)	No
Description	Short description of service	String	255 bytes(Max)	No
Cost	Cost of service	Double	8 bytes	No

Service Type

Attribute	Definition	Type	Size	Key
ServiceTypeID	Unique identifier for each service type	Int	4 bytes	Primary

Donations

Attribute	Definition	Type	Size	Key
DonationID	Unique identifier for each donation	Unsigned Long	8 bytes	Primary

ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Foreign
DonationDate	Date of donation	Date	3 bytes	No
DonationMethod	Method of payment	String	255 bytes(Max)	No
DonationAmount	Amount donated in USD	Double	8 bytes	No

Volunteers

Attribute	Definition	Type	Size	Key
VolunteerID	Unique identifier for each volunteer	Unsigned Long	8 bytes	Primary
ClientID	Unique identifier for each user	Unsigned Long	8 bytes	Foreign
RoleID	The role/work the volunteer will be tasked with	Unsigned Long	8 bytes	Foreign
AvailabilityID	Which days & times the volunteers are available	Unsigned Long	8 bytes	Foreign
StartDate	Volunteer Start Date	Date	3 bytes	No
EmergencyID	Emergency contact information for each volunteer	Unsigned Long	8 bytes	Foreign

Volunteer Availability

Attribute	Definition	Type	Size	Key
AvailabilityID	Unique identifier for each availability schedule	Unsigned Long	8 bytes	Primary
DayOfWeek	Days the volunteer are available	String	255 bytes(Max)	No
TimeSlot	Time ranges the volunteer are available	String	255 bytes(Max)	No

VolunteerRole

Attribute	Definition	Type	Size	Key

VolunteerRoleID	Unique identifier for each role	Int	4 bytes	Primary
VolunteerRoleName	Name of role	String	255 bytes(Max)	No
VolunteerRoleDescription	Short description of role	String	255 bytes(Max)	No

EmergencyContact

Attribute	Definition	Type	Size	Key
EmergencyID	Unique identifier for each emergency contact	Unsigned Long	8 bytes	Primary
FirstName	Emergency contact first name	String	255 bytes(Max)	No
LastName	Emergency contact last name	String	255 bytes(Max)	No
Relationship	Relationship of emergency contact to volunteer	String	255 bytes(Max)	No
Phone	Emergency contact phone number	String	30 bytes	No

Events

Attribute	Definition	Type	Size	Key
EventID	Unique identifier for each event	Unsigned Long	8 bytes	Primary
EventName	The name of the event	String	255 bytes(Max)	No
EventDescription	Basic description of the event and what it entails	String	255 bytes(Max)	No
EventDate	Date the event is scheduled for	Date	3 bytes	No
EventTime	The time the event is scheduled for (EST)	String	20 bytes	No
Address	Address of where the event is being hosted	String	255 bytes(Max)	No
OrganizerID	Unique identifier for each event organizer	Unsigned Long	8 bytes	Foreign

Employee

Attribute	Definition	Type	Size	Key
EmployeeID	Unique identifier for each employee	Unsigned Long	8 bytes	Primary
FirstName	First name of employee	String	255 bytes(Max)	No
LastName	Last name of employee	String	255 bytes(Max)	No
Phone	Phone number of employee	String	30 bytes	No
Email	Email of employee	String	255 bytes(Max)	No
StartDate	Date the employee begun working	Date	3 bytes	No
EmployeeRoleID	Position/Role of employee	Unsigned Long	8 bytes	Foreign
Username	Username of employee	String	255 bytes(Max)	No
PasswordHash	Password of employee	String	255 bytes(Max)	No
Salary	Defined salaries of each employee	String	255 bytes(Max)	No

EmployeeRole

Attribute	Definition	Type	Size	Key
EmployeeRoleID	Unique identifier for each employee role	Int	4 bytes	Primary
EmployeeRoleName	Name for each employee role	String	255 bytes(Max)	No
EmployeeRoleDescription	Short description for each role	String	255 bytes(Max)	No

EventOrganizer

Attribute	Definition	Type	Size	Key
OrganizerID	Unique identifier for each event organizer	Unsigned Long	8 bytes	Primary
EmployeeID	Unique identifier for each employee	Unsigned Long	8 bytes	Foreign

Blog

Attribute	Definition	Type	Size	Key
BlogID	Unique identifier for each blog post	Unsigned Long	8 bytes	Primary
EmployeeID	Unique identifier for each employee	Unsigned Long	8 bytes	Foreign
Description	Description of each blog post	String	255 bytes(Max)	No
Date	Date of when each blog post is posted	Date	3 bytes	No

ContactRequest

Attribute	Definition	Type	Size	Key
RequestID	Unique identifier for each contact request submitted			

ChatBot

Attribute	Definition	Type	Size	Key
ChatbotID	The unique identifier for the chatbot	Int	4 bytes	Primary
ChatHistoryID	Unique identifier for each chat history	Unsigned Long	8 bytes	Foreign
EmployeeID	Unique identifier for each employee	Unsigned Long	8 bytes	Foreign

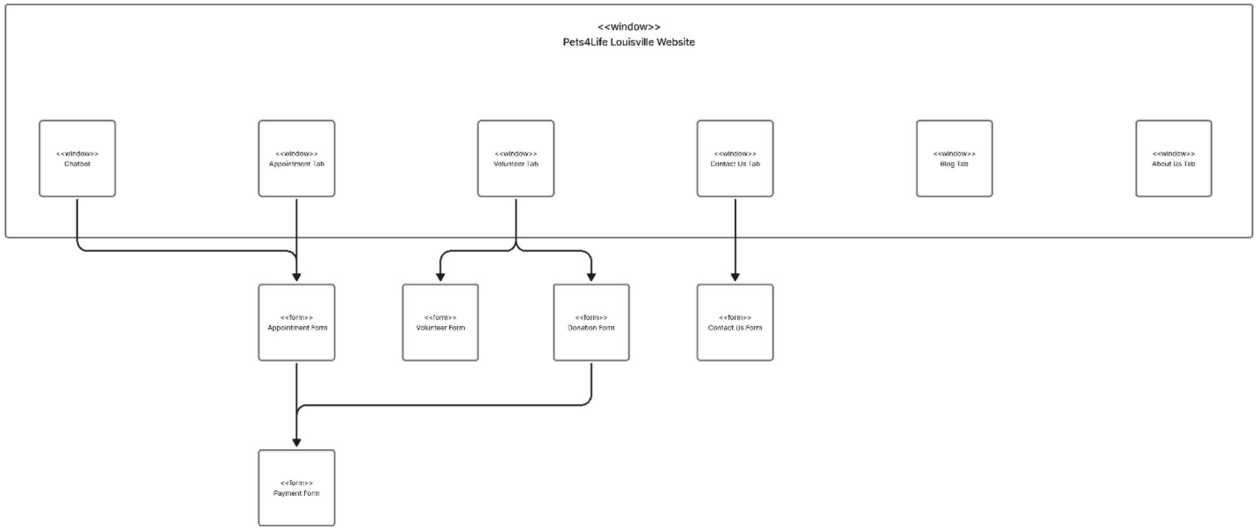
ChatBotChatHistory

Attribute	Definition	Type	Size	Key
ChatHistoryID	Unique identifier for each chat history	Unsigned Long	8 bytes	Primary
Messages	What is contained within the string of messages	String	255 bytes(Max)	No
Description	A short summary of the conversation with the chatbot	String	255 bytes(Max)	No

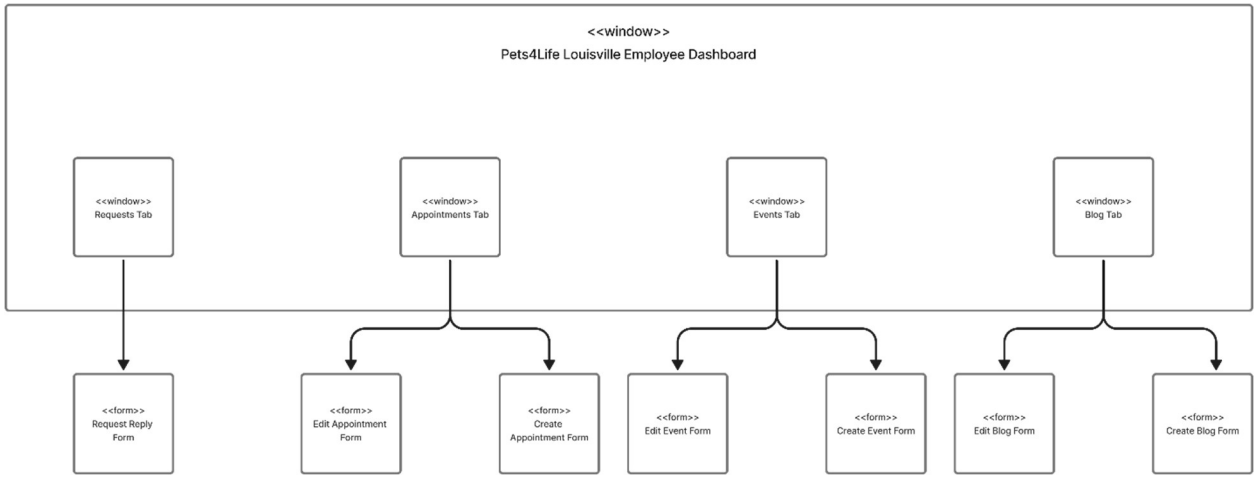
3. User Interface Design

3.1 User Interface Navigation Diagram

3.1.A Pets4Life Louisville Website



3.1.A Pets4Life Louisville Employee Dashboard



3.2 User Interface Diagram Narrative

Pets4Life Louisville Website User Interface Navigation Diagram is a high-level overview of the navigation flow for the Pets4Life Louisville website. Users primarily navigate through the website by clicking on the numerous tab items on the top navigation bar. Throughout the entire website, the navigation hierarchy is mostly one screen deep, with most leading to a form where user information is needed. Appointments and donations will lead the user to a payment screen. Some screens, such as the

About Us Tab and Blog Tab, are read only where content from Pets4Life Louisville's employees can be seen.

Pets4Life Louisville Employee Dashboard User Interface Navigation Diagram expresses a very similar user interface to the website. The key difference is the tabs allows for the create, read, update and delete actions for the four key components:

User Requests

Appointments

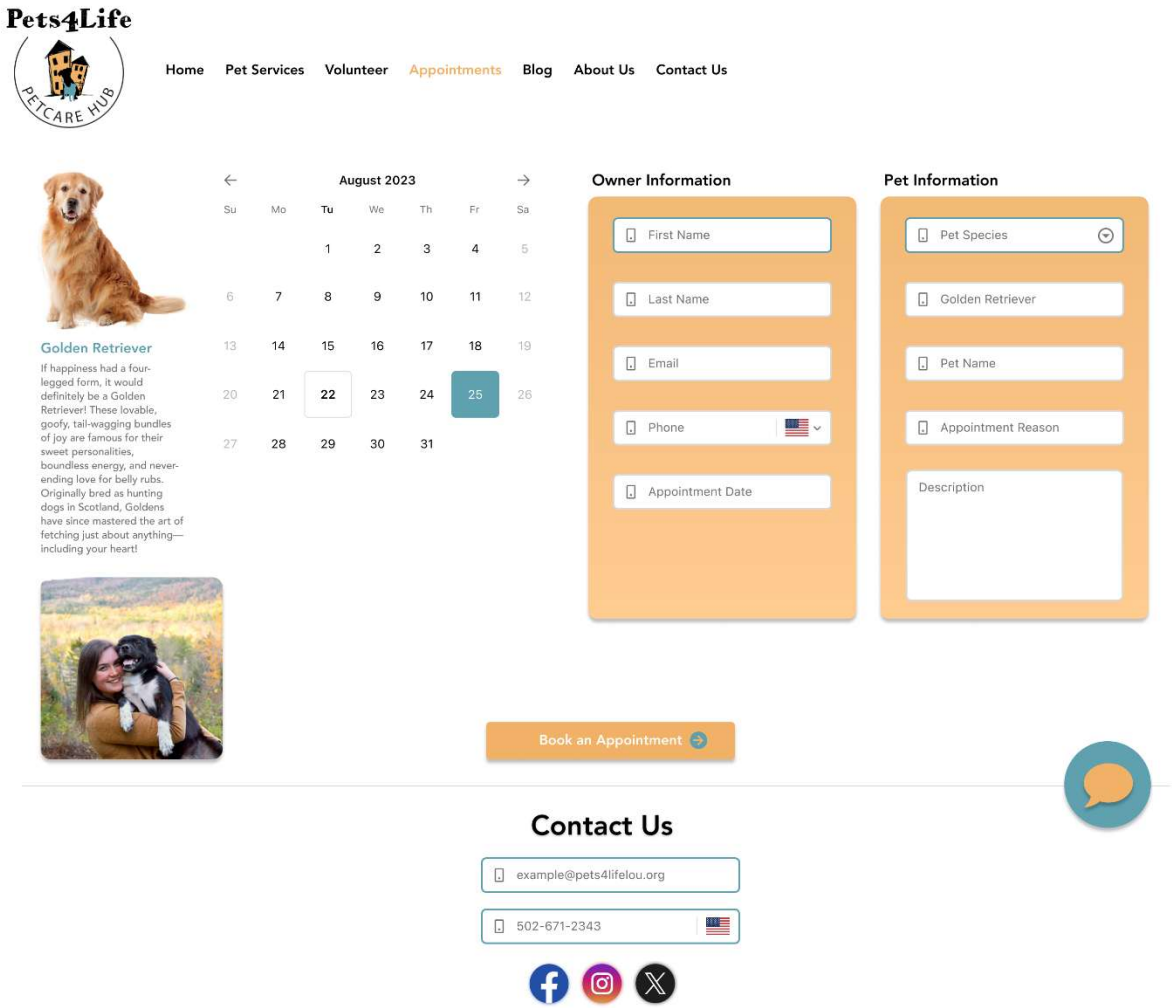
Volunteering Events

Blog Posts

3.3 Screen Layouts

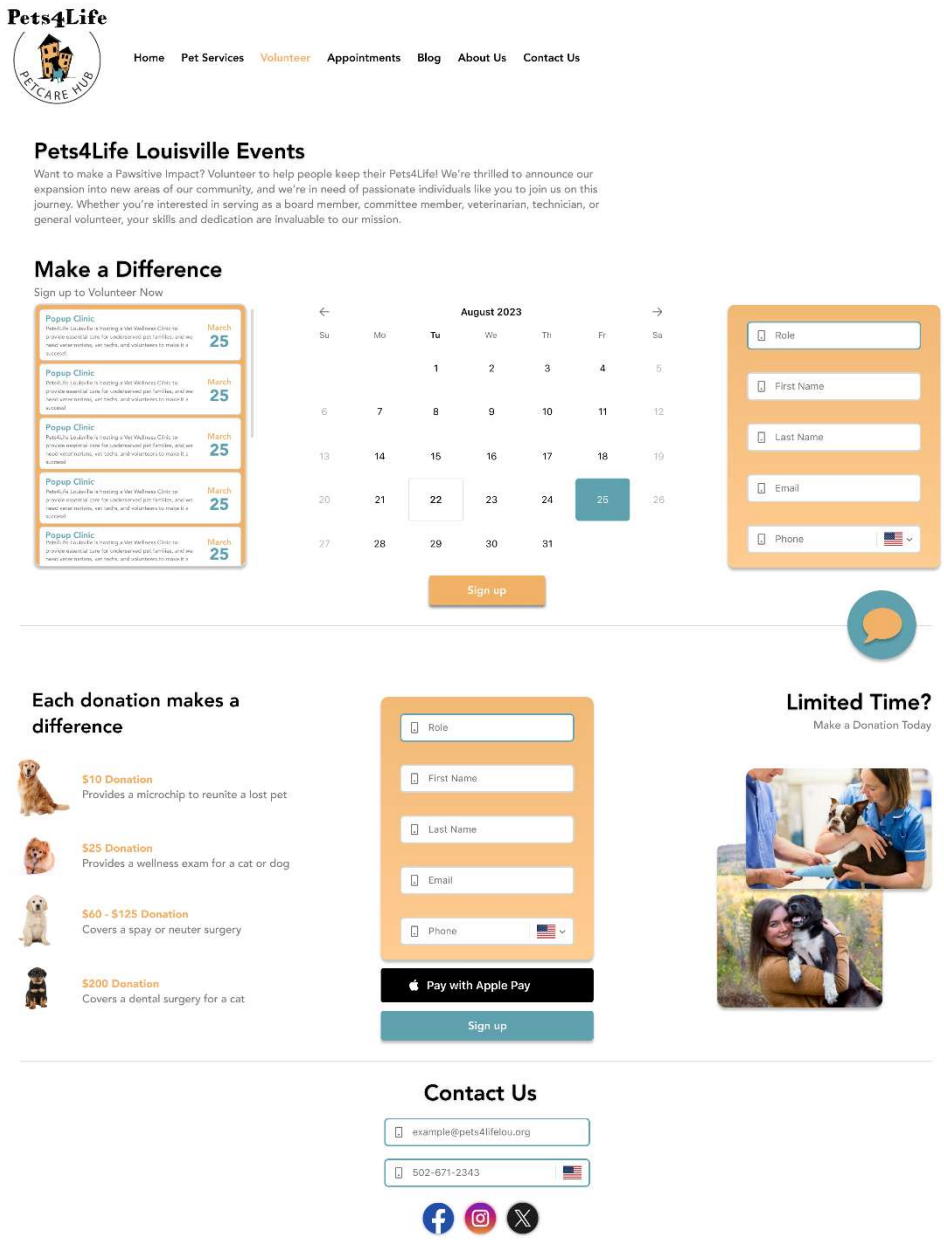
3.3.A Pets4Life Louisville Website

Appointment Tab



The Appointment Tab on the Pets4Life website is designed to facilitate efficient scheduling of various pet services, including immunization, spay/neuter, and surgery appointments. The screen layout incorporates an intuitive form for capturing both owner and pet information, ensuring all necessary details are provided for the appointment. A dynamic calendar allows users to select and view available dates easily, streamlining the booking process. This setup not only enhances user experience by making appointment scheduling straightforward but also ensures that all required information is relayed accurately to the administrators, supporting high-risk service appointments effectively. The integration of clear, accessible forms and a responsive calendar directly supports the primary use cases by allowing users to specify the type of service needed, thereby optimizing administrative preparation and customer satisfaction.

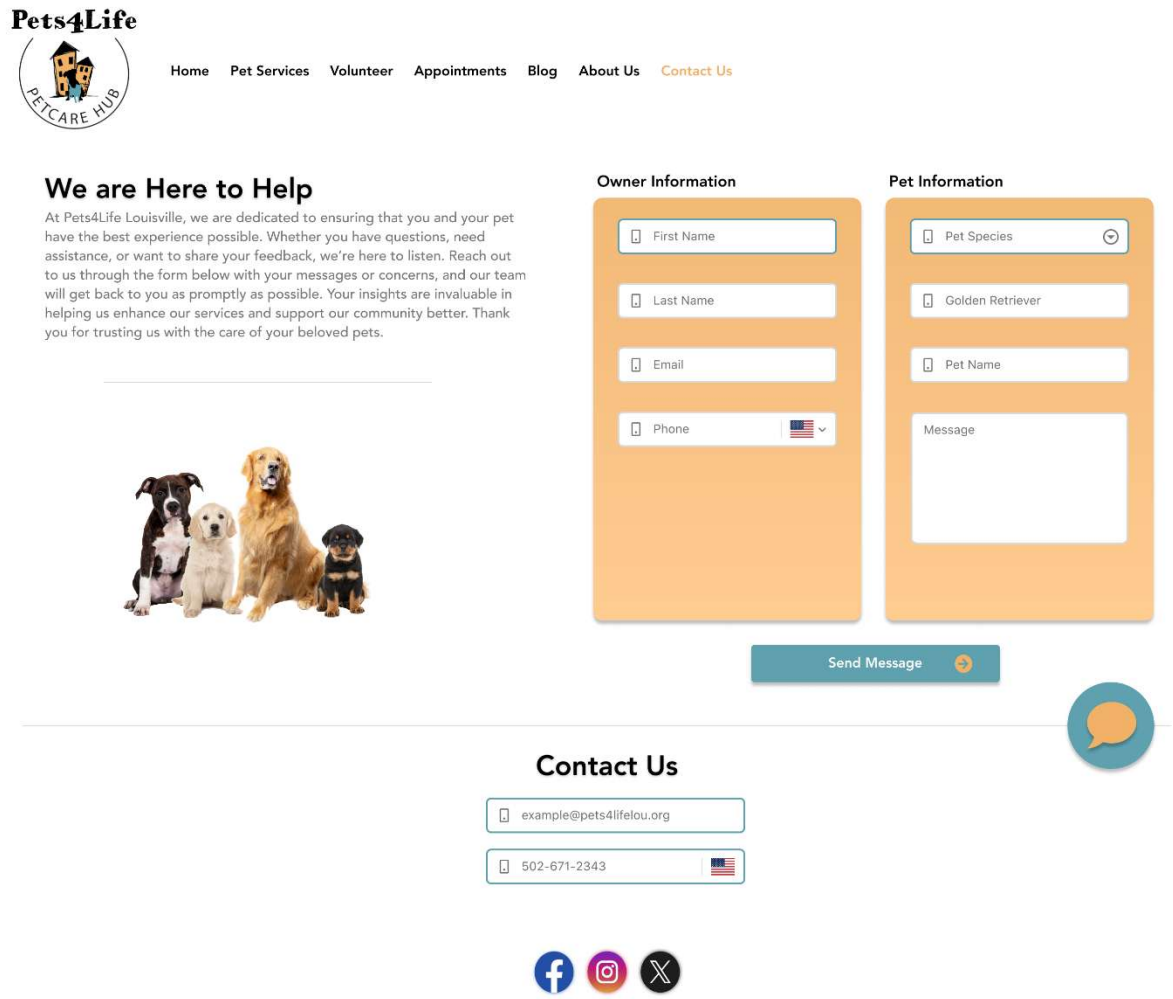
Volunteer Tab



The Volunteer Tab on the Pets4Life website is designed to cater to user participation in philanthropy and community service, closely aligning with use cases such as Donating Pre-Determined Amounts, Donating Custom Amounts, and Applying for Volunteer Work. This streamlined interface enhances user experience by offering intuitive interactions for making financial contributions and signing up for volunteer opportunities.

For donations, the tab features clearly delineated sections where users can choose between predetermined amounts, each describing the specific services or benefits that the donation will support, such as microchipping a lost pet or covering a wellness exam. For users who prefer a more tailored approach, an option to specify a custom donation amount is readily accessible, ensuring flexibility to contribute as they see fit.

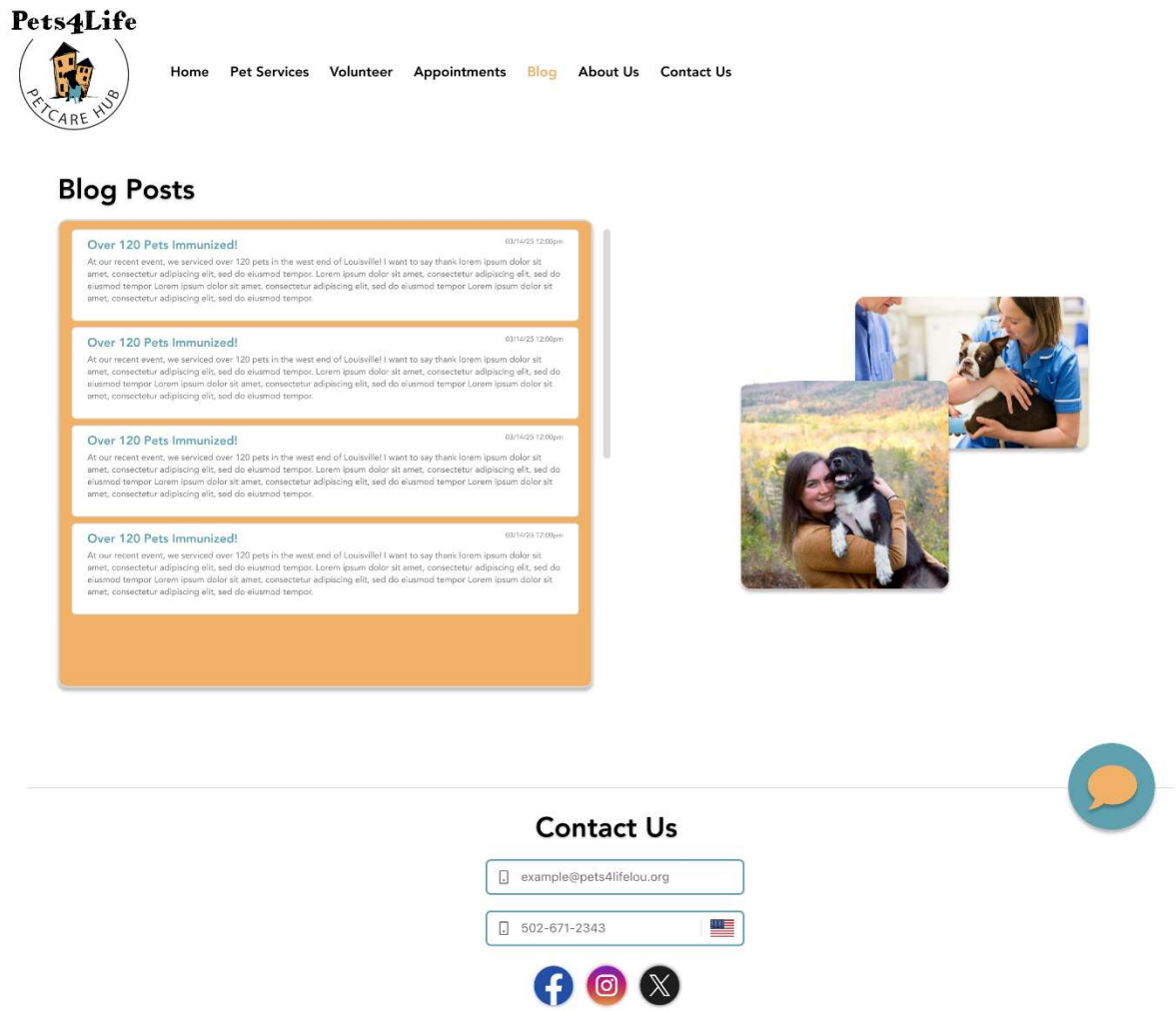
Contact Us Tab



The Contact Us Tab on the Pets4Life Louisville website is designed to support user engagement through straightforward communication channels. This design is particularly vital for realizing the use cases of Creating User Requests and Generating Tickets (CRM). For Creating User Requests, the tab features an intuitive form where users can input their personal and pet information, along with a message detailing their request or concern. This form is structured to collect all necessary details that would allow the organization to effectively respond to the user. The simplicity of the form encourages user participation by making it easy to reach out for help or information, thereby enhancing user satisfaction and trust in the organization.

The information submitted through this form is automatically integrated into the CRM system, Jira. This system generates tickets based on user submissions, which are then made visible to administrators. This automation streamlines the process of ticket generation from various inputs, including direct form submissions and interactions on social media platforms.

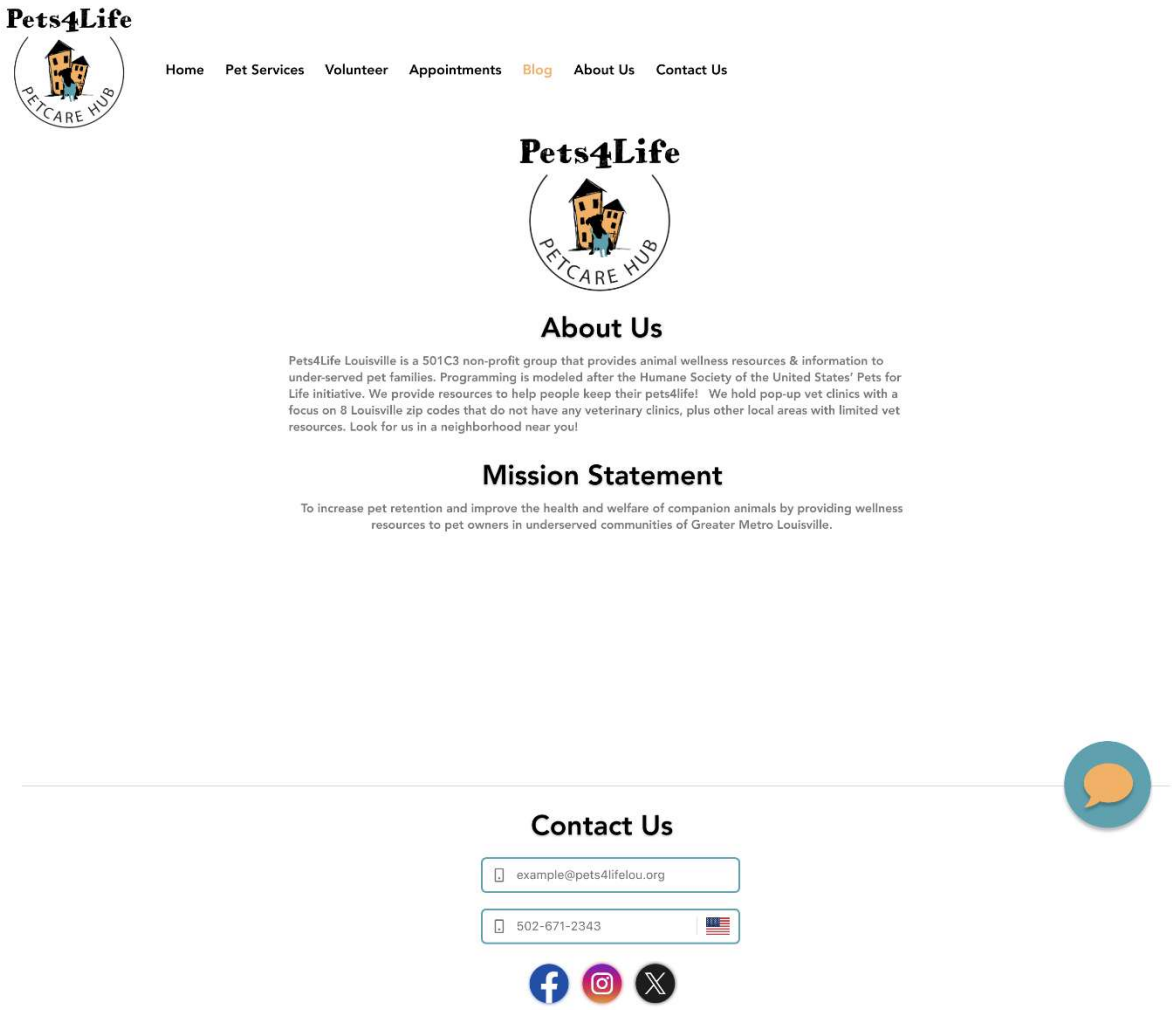
Blog Tab



The Blog Tab on the Pets4Life Louisville website gives users easy access to a variety of content, including articles, photos, videos, and updates from the organization. This tab fulfills the use case of Using the Blog, where the primary actor is the User who seeks to engage with and consume informational and educational content provided by Pets4Life Louisville.

The layout of the Blog Tab features a clean, organized design with multiple posts displayed on a single page. Each blog entry includes a headline, a brief excerpt, and a timestamp, alongside appealing images that invite further reading. This design allows users to quickly scan through the latest updates or dive deeper into specific topics of interest. The user-friendly interface enhances the browsing experience, making it straightforward for users to navigate through different posts.

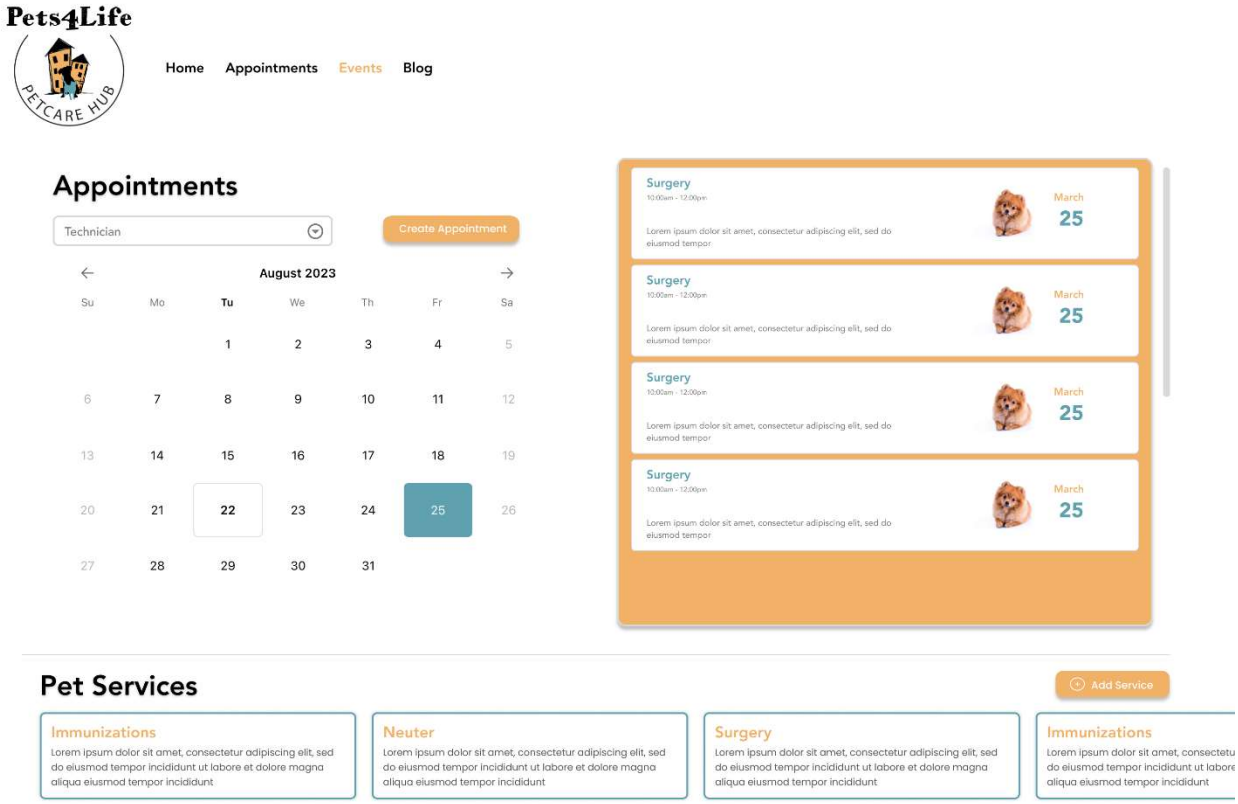
About Us Tab



The About Us Tab on the Pets4Life website is designed to provide users with a clear and informative overview of the organization's mission, history, and objectives. This tab directly addresses the use case of Visiting 'About Us' Tab, where the primary actor, the user, seeks to understand more about the organization's purpose and the context of its services.

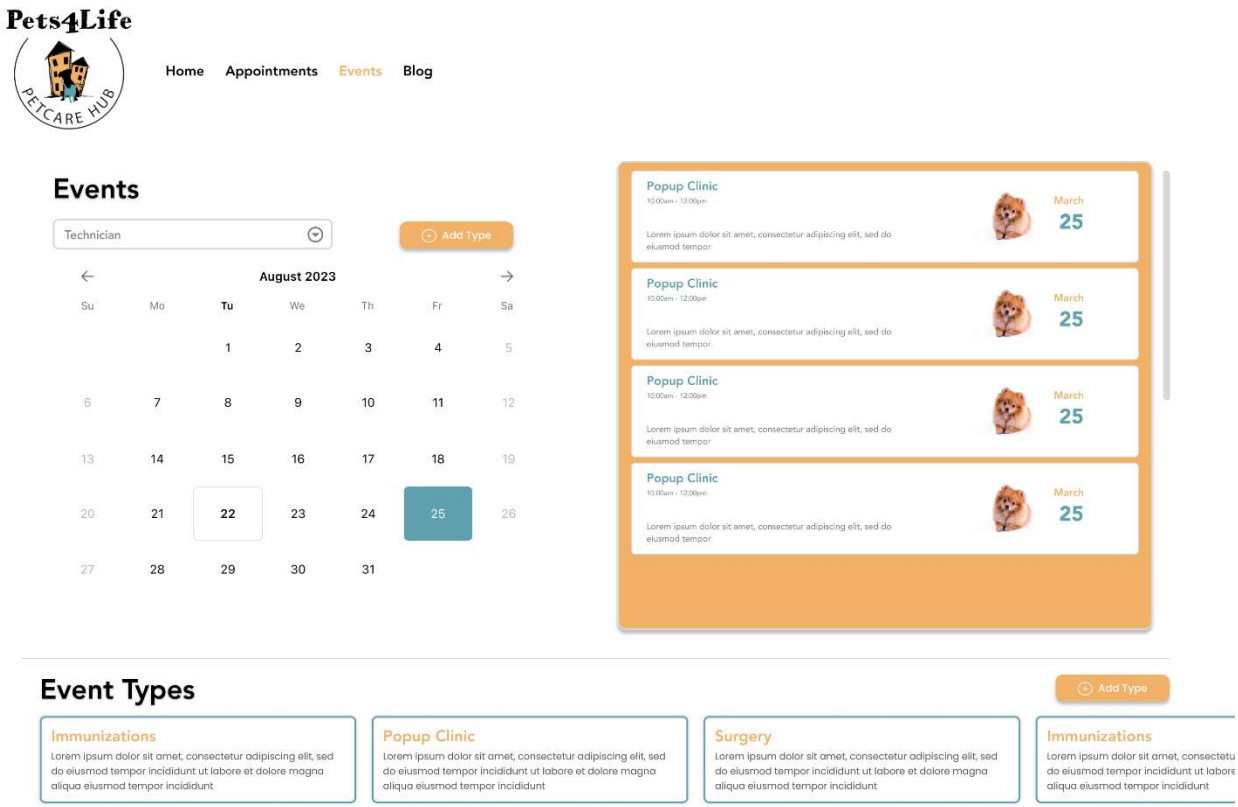
3.3.B Pets4Life Louisville Employee Dashboard

Dashboard Appointment Tab



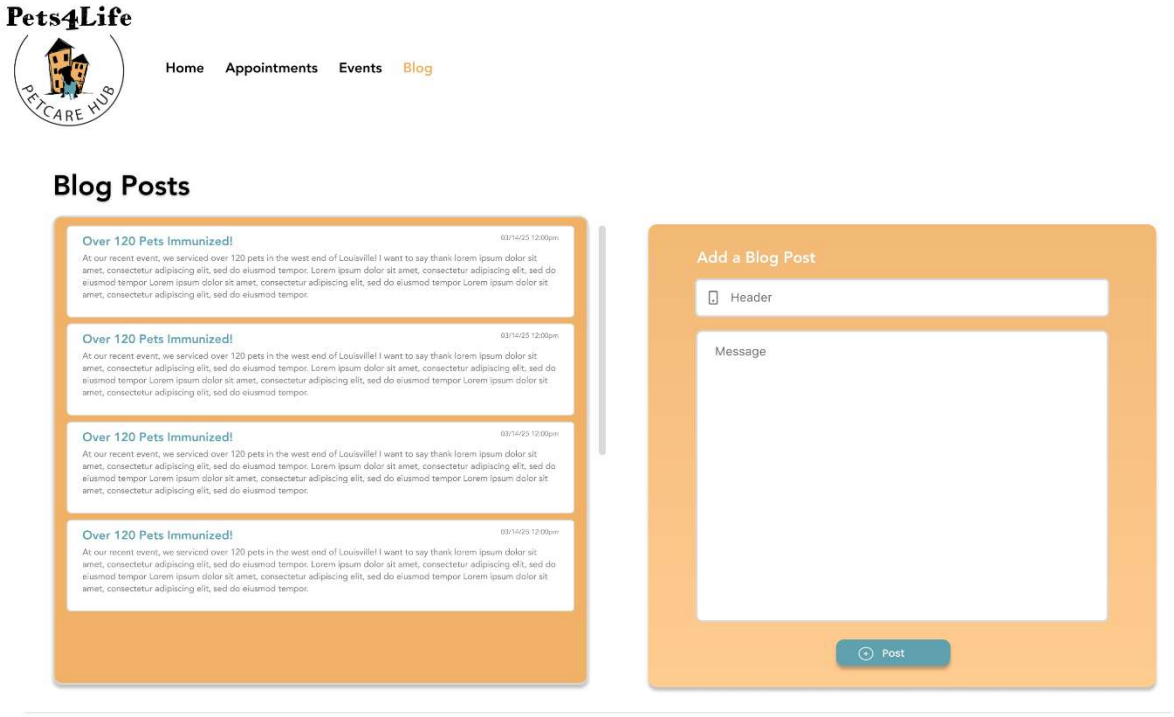
The Employee Dashboard's Appointments Tab on the Pets4Life Louisville Employee Dashboard website is a critical tool for administrators, designed to streamline the management of various appointment-related tasks. It effectively addresses several use cases, enhancing administrative control and efficiency. For Customizing Appointment Types, the dashboard allows administrators to add, edit, or remove appointment types, ensuring the services offered are up-to-date and relevant. In terms of Managing Appointments, the interface displays a comprehensive calendar view of all appointments, with functionalities to reschedule or cancel directly from the dashboard, facilitating efficient operation and responsive customer service. Additionally, **Viewing User Contact Information** is seamlessly integrated, allowing administrators to access contact details within the appointment entries for easy communication with clients. This integrated approach ensures the dashboard is not only a tool for managing appointments but also a central hub for enhancing communication and adapting service offerings based on client needs and organizational changes.

Dashboard Volunteer Tab



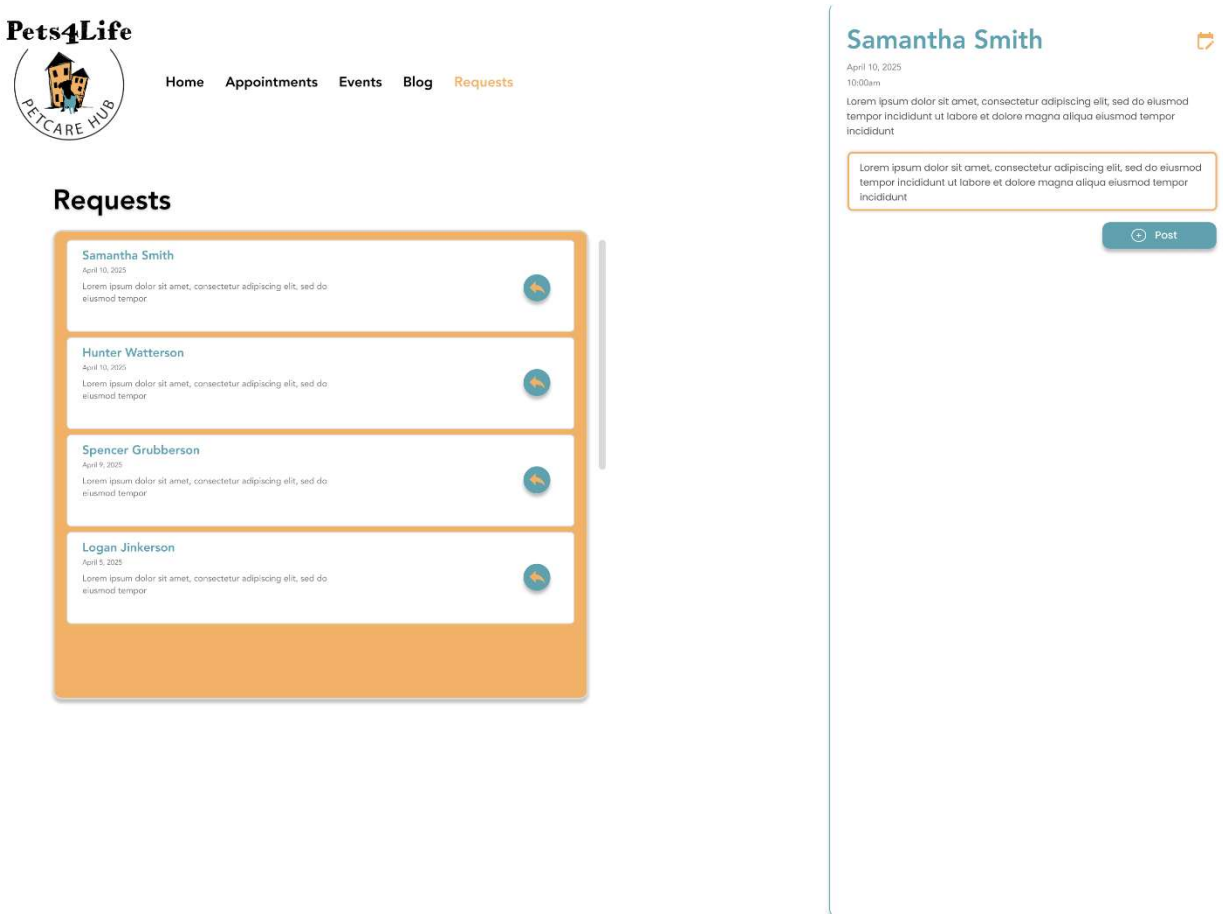
The Volunteer Dashboard on the Pets4Life website, particularly in the Events section, is an advanced tool designed to enhance the efficiency of administrators managing volunteer activities. It supports critical functionalities such as viewing all registered events, managing participant lists, and accessing relevant contact details, along with capabilities to create, modify, or cancel events. This interface includes an interactive calendar and options for detailed event management, providing administrators with the necessary tools to effectively organize, track, and adjust volunteer events. The dashboard ensures that administrators can swiftly respond to changes, maintain effective communication with volunteers, and adapt to community needs, thereby optimizing event planning and volunteer management processes.

Dashboard Blog Tab



The Dashboard Blog Tab on the Pets4Life website is effectively designed to facilitate the management of blog content by administrators, aligning seamlessly with use case Customizing Blog Posts. This tab enables administrators to perform critical content management functions such as adding new blog posts, editing existing entries, or deleting them as necessary.

Dashboard Requests Tab



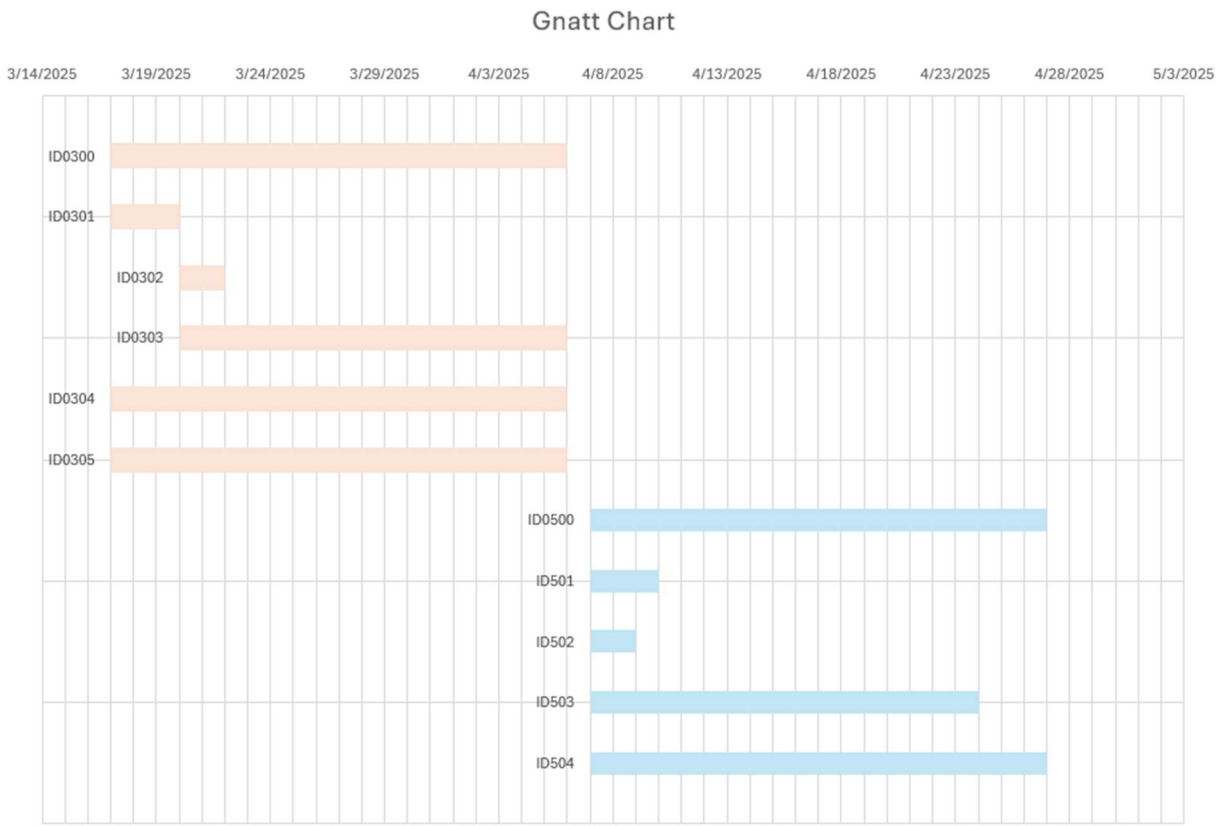
The Dashboard Requests Tab on the Pets4Life website is a crucial feature designed to help administrators efficiently manage and respond to user inquiries, aligning perfectly with use case Respond to User Requests. This tab displays a clear and organized list of all incoming requests from the 'Contact Us' tab, allowing administrators to see each request immediately, with details such as the requester's name, the date of the request, and a brief snippet of the message.

Administrators can select any request to view it in more detail on an adjacent pane, which provides the full message content and allows for direct response. This feature ensures that all user inquiries are addressed in a timely and organized manner, enhancing customer service and satisfaction. The interface is designed to be intuitive, with user requests clearly listed and easily accessible, minimizing the time and effort required to manage communications. This streamlined process not only optimizes administrative tasks but also supports maintaining high standards of responsiveness and engagement with the Pets4Life community.

4. Gantt Chart

4.1 Gantt Chart Diagram

- Excel sheet link

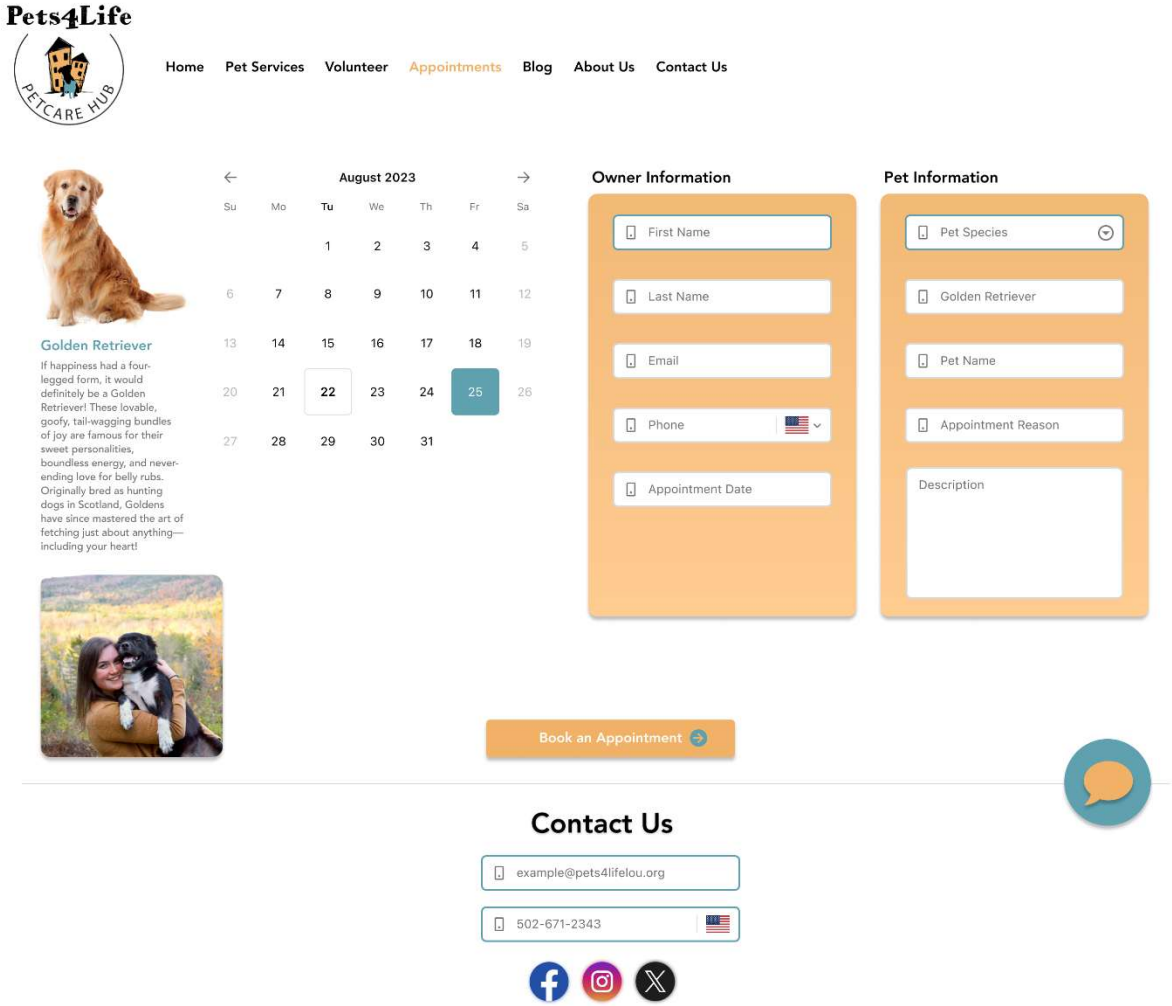


Task ID	Task Name	Start Date	End Date	Duration	Task Responsibility	Dependencies
ID0300	I3 Document	3/17/2025	4/6/2025	20	Spencer Taylor, Logan Riedley, Hunter Watring, Gerrit Kloek	No
ID0301	Use Cases	3/17/2025	3/20/2025	3	Spencer Taylor, Logan Riedley, Hunter Watring, Gerrit Kloek	No
ID0302	Trace Matrix	3/20/2025	3/22/2025	2	Logan Riedley, Hunter Watring	ID0301
ID0303	Use Case Diagram	3/20/2025	4/6/2025	17	Spencer Taylor, Logan Riedley, Hunter Watring, Gerrit Kloek	ID0301
ID0304	Gnatt Chart	3/17/2025	4/6/2025	20	Spencer Taylor	No
ID0305	Use Case Prototype	3/17/2025	4/6/2025	20	Logan Riedley	No
ID0500	I5 Document	4/7/2025	4/13/2025	6	Spencer Taylor, Logan Riedley, Hunter Watring, Gerrit Kloek	No
ID501	Database Design	4/7/2025	4/12/2025	5	Hunter Watring, Gerrit Kloek	No
ID502	Database Definition	4/7/2025	4/12/2025	5	Hunter Watring, Gerrit Kloek	No
ID503	Class Diagram	4/7/2025	4/12/2025	5	Spencer Taylor	No
ID504	User Inteface Design	4/7/2025	4/12/2025	5	All	No

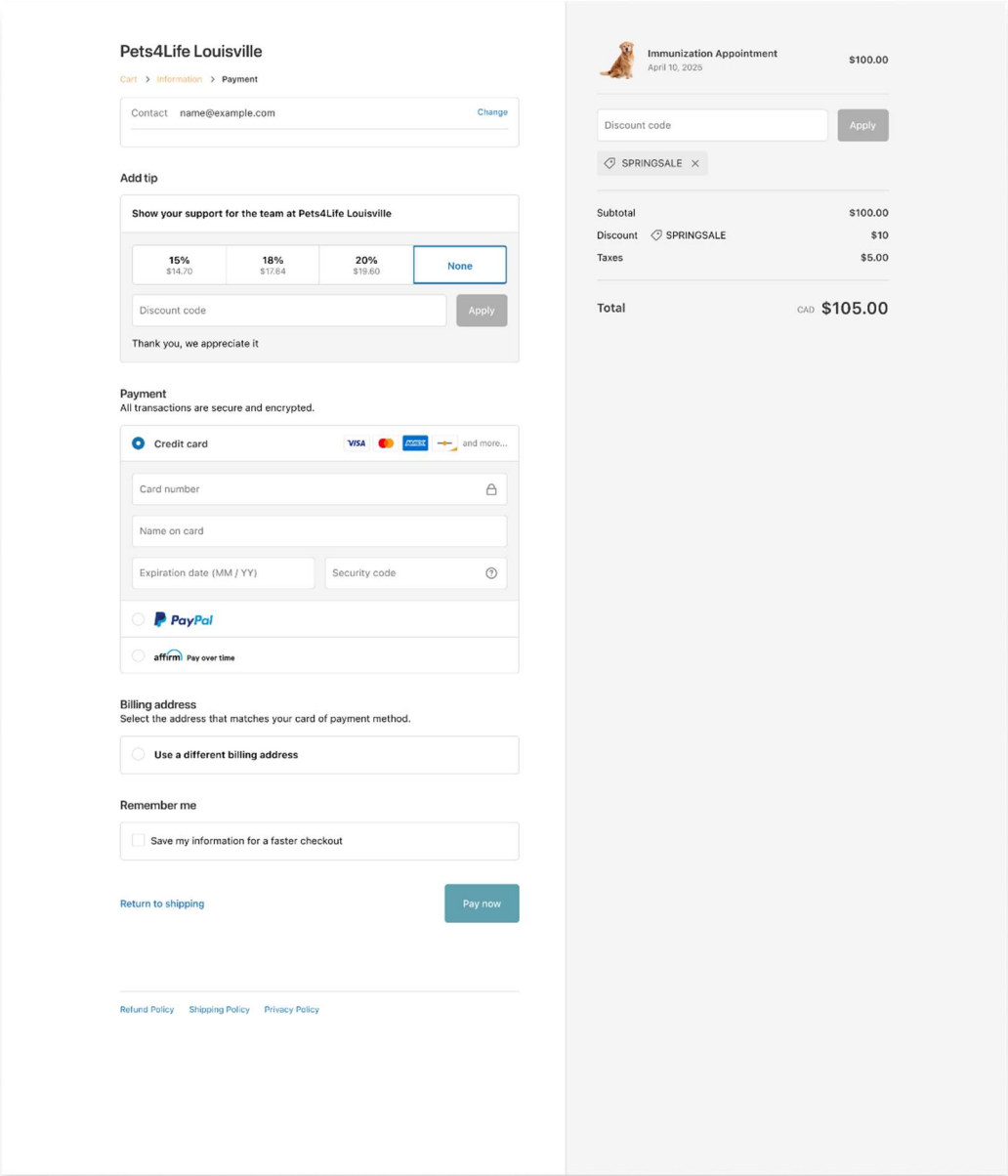
Key
I3
I5

4. User Interface Prototypes

Use Case 1 – 4, 12, 26: Book Appointments



Use Case 5: Processing Online Payments



Use Case 7 – 8, 14 -15: Volunteer Events and Donations

Pets4Life

PETCARE HUB

Home Pet Services Volunteer Appointments Blog About Us Contact Us

Pets4Life Louisville Events

Want to make a Pawsitive Impact? Volunteer to help people keep their Pets4Life! We're thrilled to announce our expansion into new areas of our community, and we're in need of passionate individuals like you to join us on this journey. Whether you're interested in serving as a board member, committee member, veterinarian, technician, or general volunteer, your skills and dedication are invaluable to our mission.

Make a Difference

Sign up to Volunteer Now

Popup Clinic March 25

Popup Clinic March 25

Popup Clinic March 25

Popup Clinic March 25

Popup Clinic March 25

August 2023

Su	Mo	Tu	We	Th	Fr	Sa
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Book

Role

First Name

Last Name

Email

Phone

Each donation makes a difference

$10 Donation
Provides a microchip to reunite a lost pet

$25 Donation
Provides a wellness exam for a cat or dog

$60 - $125 Donation
Covers a spay or neuter surgery

$200 Donation
Covers a dental surgery for a cat

Role

First Name

Last Name

Email

Phone

Pay with Apple Pay

Sign up

Limited Time?

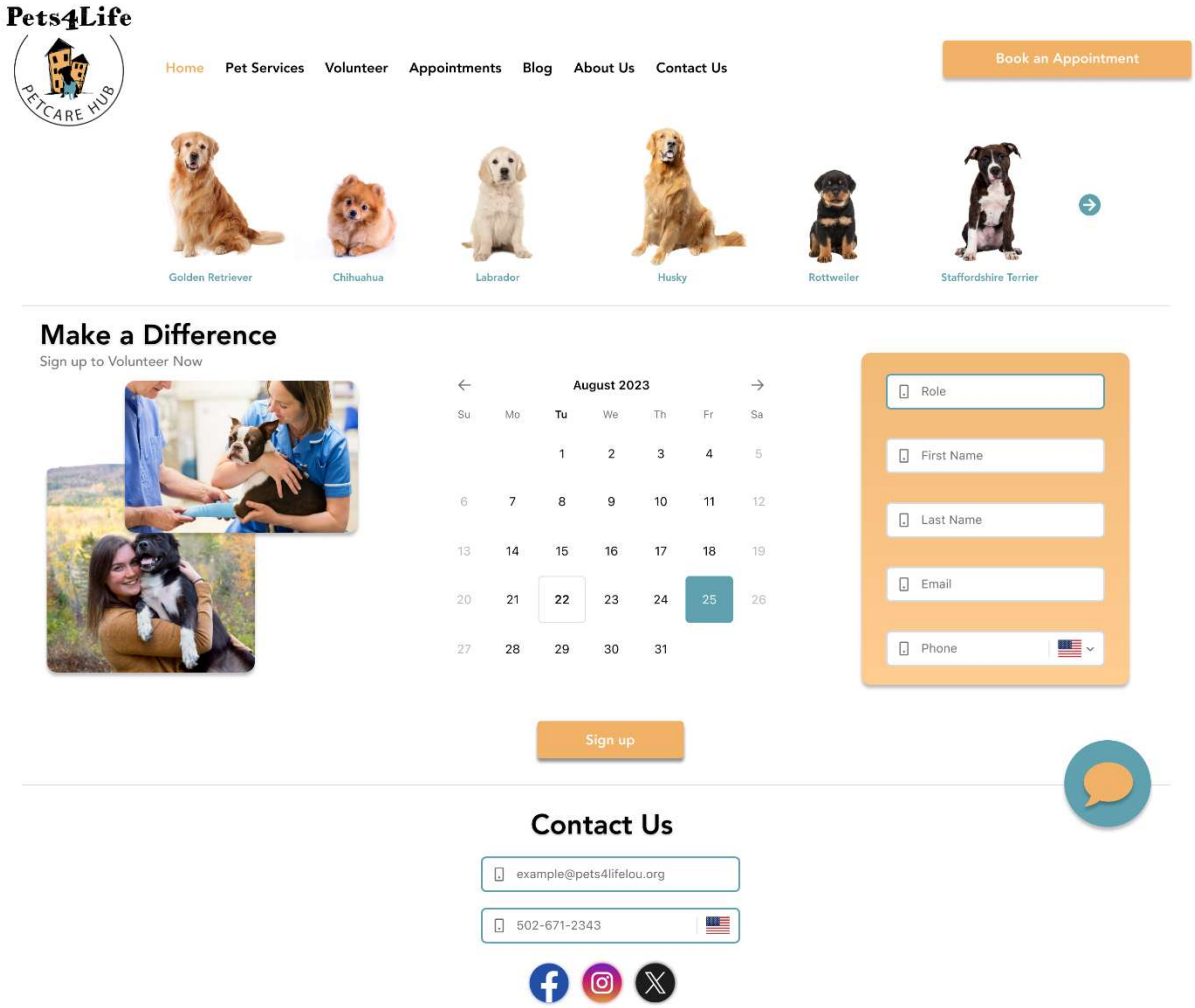
Make a Donation Today

Contact Us

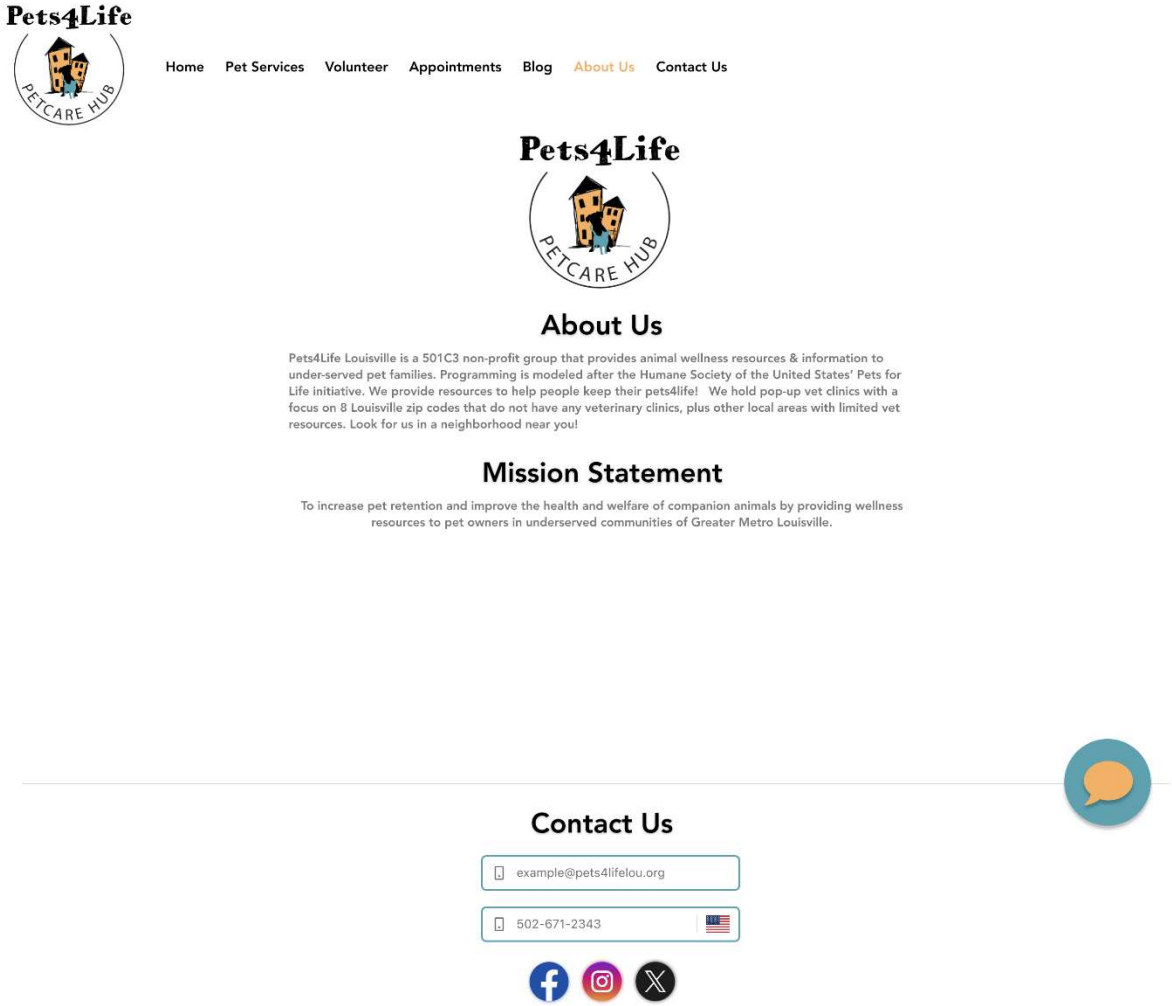
example@pets4lifelou.org

502-671-2343

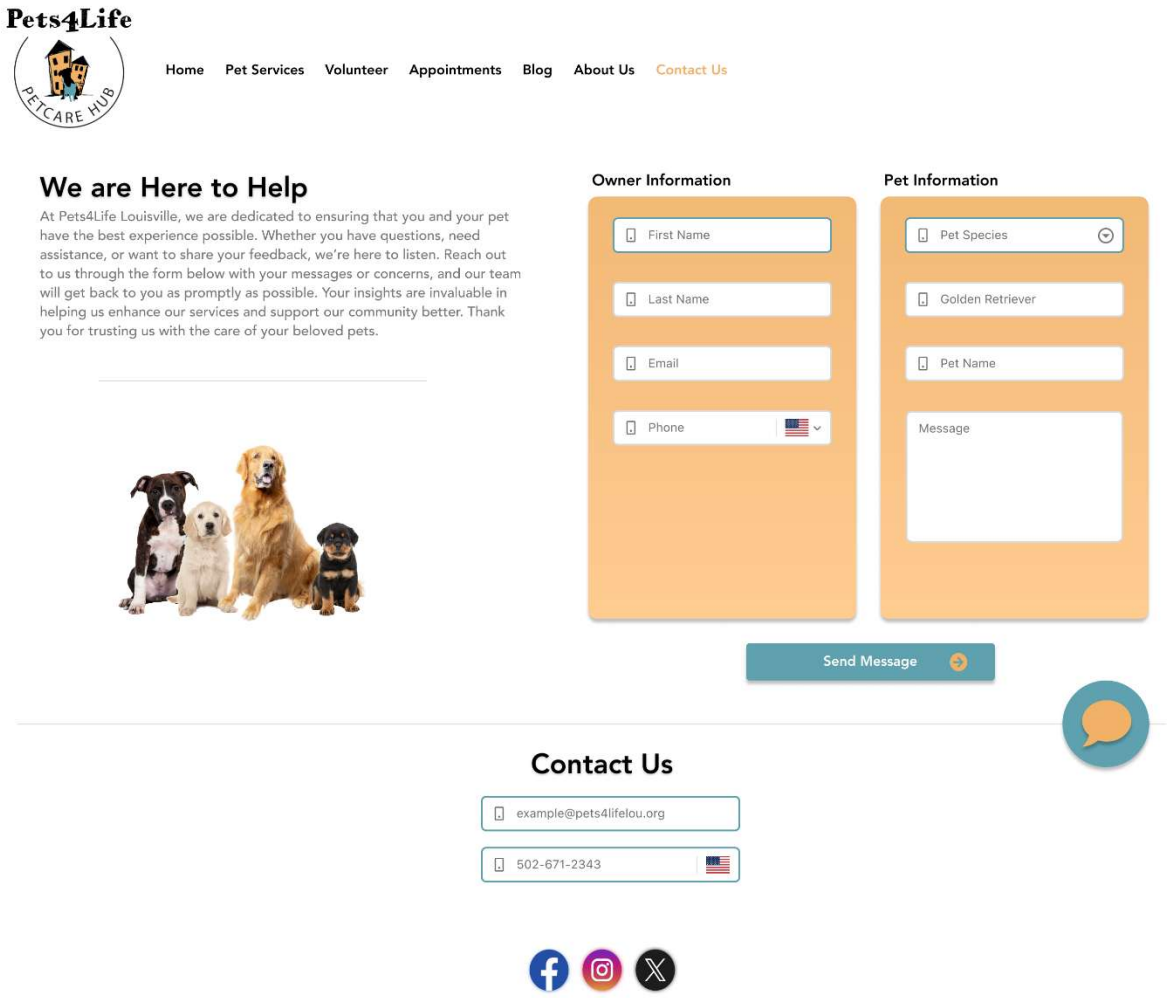
Use Case 10 – 11: Chatbot



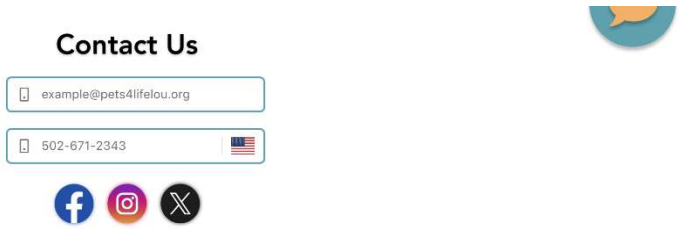
Use Case 1**1**: Visiting ‘About Us’ Tab



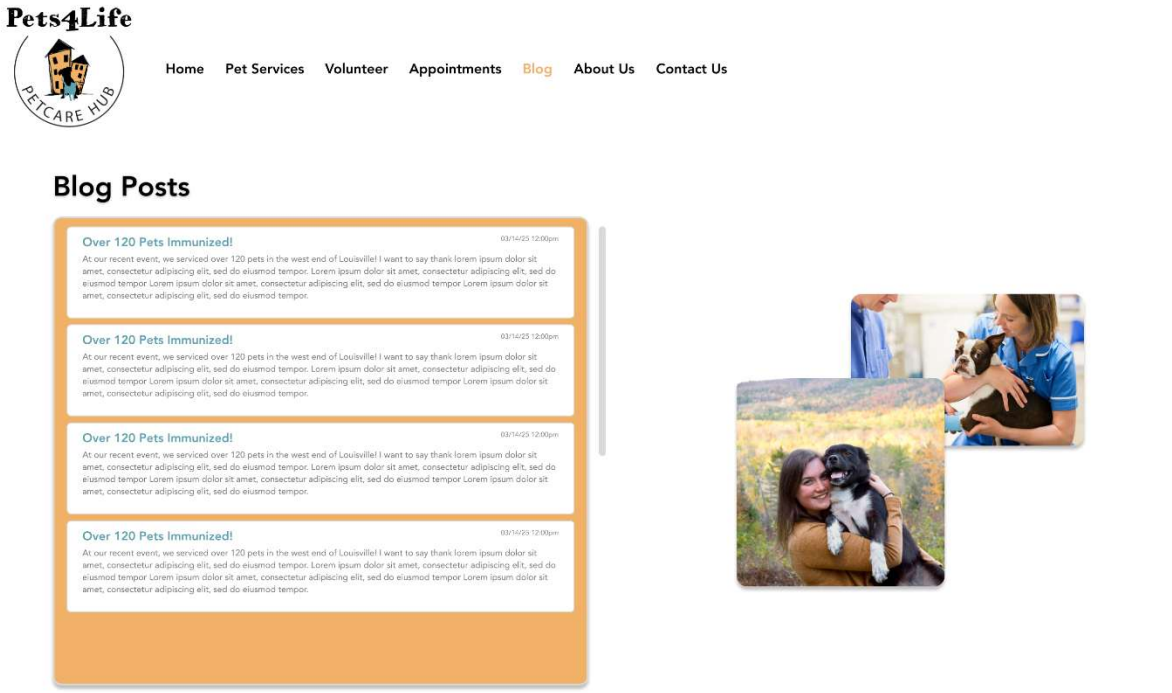
Use Case 13, 17: User Requests



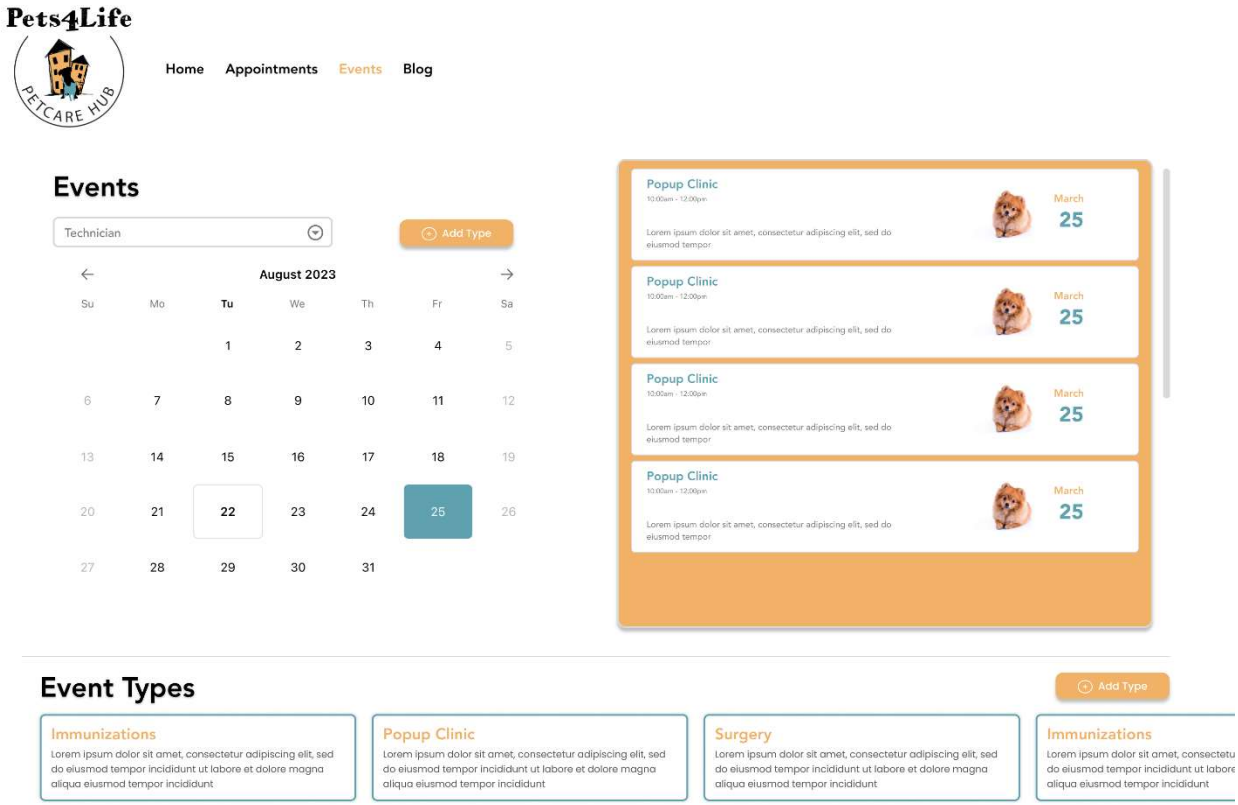
Use Case 16, 28: Viewing Administrator Contact Information and Social Media Links



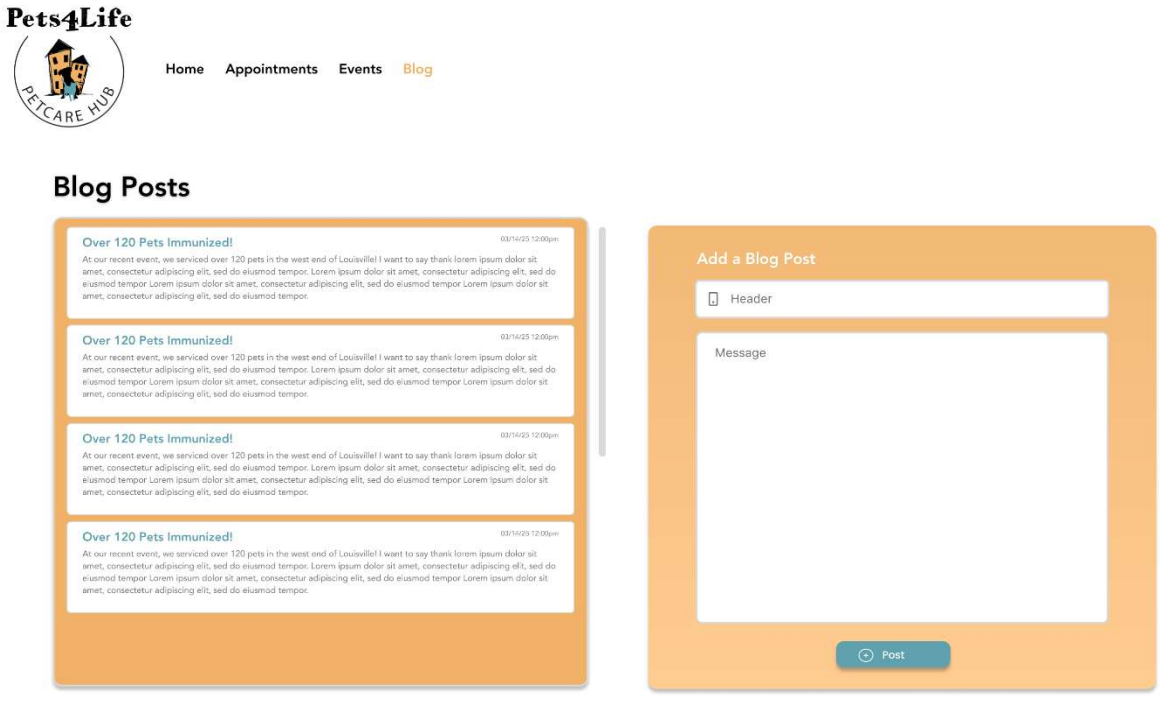
Use Case 18: Using the Blog



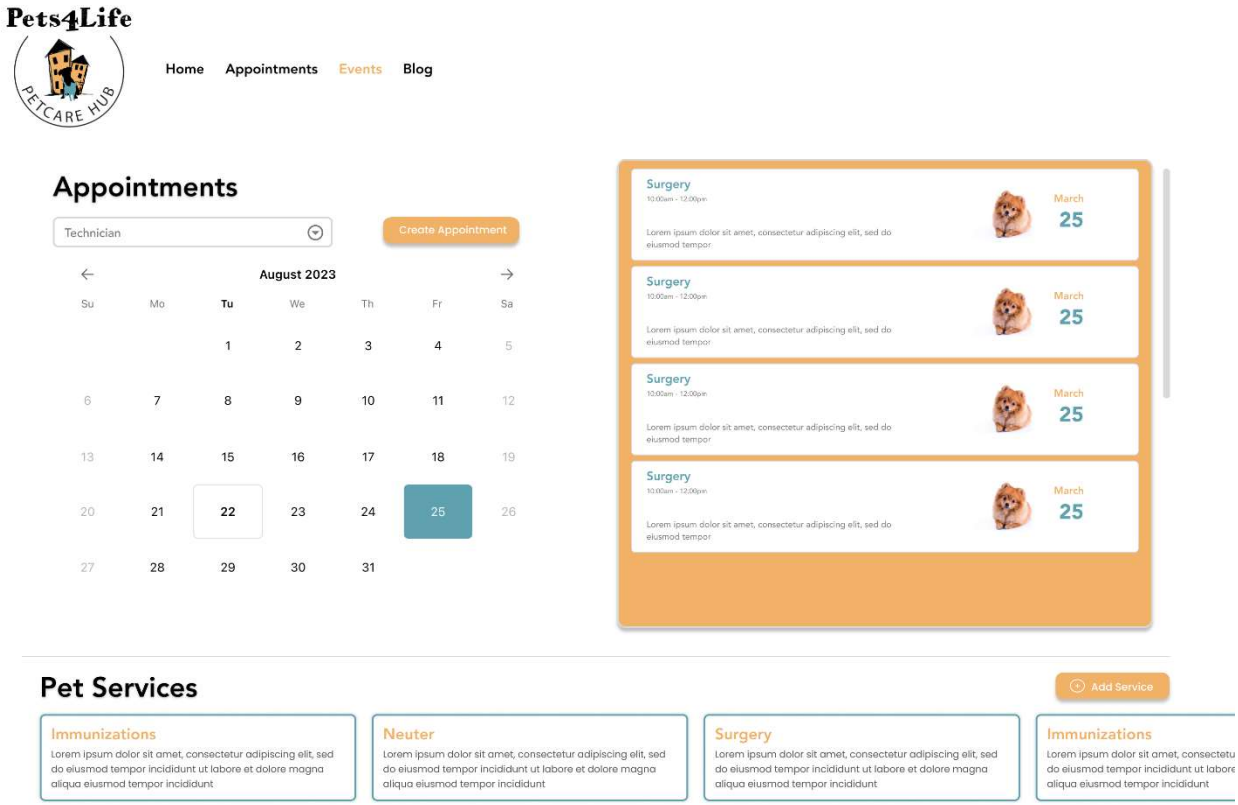
Use Case 19 - 20: Volunteering Events



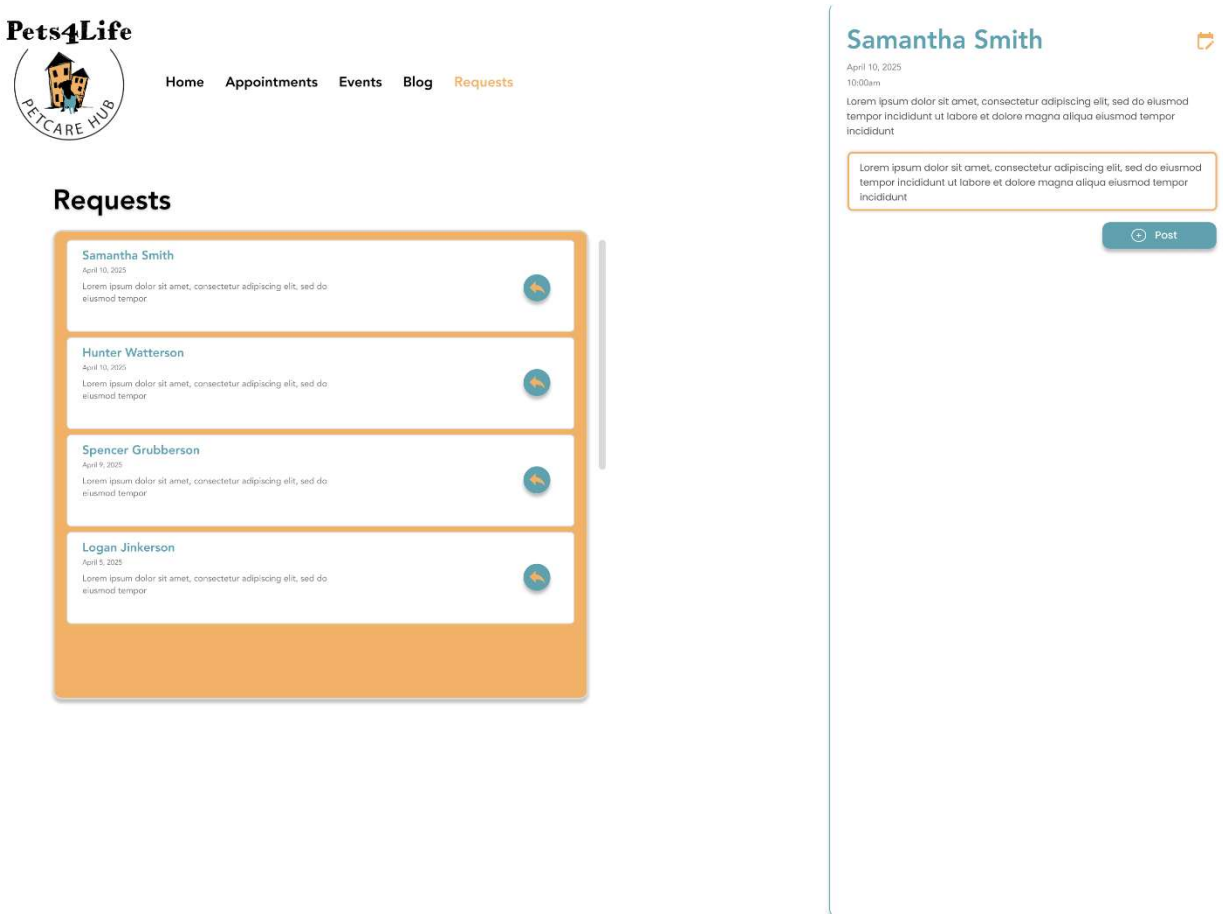
Use Case 21: Customizing Blog Posts



Use Case 22 - 24: Appointment Management



Use Case 25: Respond to User Requests


Pets4Life
PETCARE HUB
Home
Appointments
Events
Blog
Requests
Requests
Samantha Smith
April 10, 2025
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
Hunter Watterson
April 10, 2025
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
Spencer Grubberson
April 9, 2025
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
Logan Jinkerson
April 5, 2025
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
Samantha Smith
April 10, 2025
10:00am
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua eiusmod tempor incididunt
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